

Sparse, Low-Rank, and Deep: A Compressed-Sensing Framework for Music Source Separation

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Abstract. We cast music vocal/accompaniment separation as a compressed-sensing problem on the short-time Fourier spectrogram and build five separators, each tied to one lecture of ENGS 109: Robust PCA (L13), class-conditional NMF (L15), K-SVD dictionaries with sparse-representation classification (L14, L8), a Deep Sparse Coding Network mask predictor (L16), and Multiple-Measurement-Vector joint sparsity for stereo (L12). Our central contribution is methodological: Chin’s SparseNet, previously evaluated only on image classification, is here adapted to spectrogram-domain dense regression by treating each composite sparse-coding module’s output as a learned soft mask, trained end-to-end through unrolled non-negative FISTA. On the MUSDB18 sample benchmark we observe a clean progression in median vocal SI-SDR from -3.6 dB (unsupervised Robust PCA) to $+2.7$ dB (discriminative K-SVD), while SparseNet matches K-SVD on vocals ($+2.5$ dB) and attains the best accompaniment quality ($+8.2$ dB) among all learned masks, approaching the $+15.5$ dB ideal-ratio oracle ceiling. We also report a negative-but-informative result: the learned overcomplete dictionaries have coherence $\mu \approx 0.85$, so the exact-recovery guarantees from L6–L7 are loose, yet empirical separation far exceeds the worst-case bound. All code, trained models, and playable audio are released.

1 Introduction

Music source separation recovers individual stems (vocals, drums, bass, other) from a finished stereo mixture. It is the cleanest real-world instance of the project topic suggested in this course, “blind source separation: isolate instrumental music from vocal component,” and it sits on the critical path of a product I have been building for three years.

Three acts under one advisor. This is the third time I study audio under Peter Chin. My undergraduate honors thesis built Token U-Net, a 1.08-billion-parameter encoder–decoder on EnCodec [1] tokens for full-mix music restoration. My MS thesis built MODULO/Mozart AI, a co-creative DAW whose stem-separation stage is outsourced to the ElevenLabs cloud API and scores roughly 35 dB SI-SDR below open-source Demucs [2] on internal tests. ENGS 109 supplies a third methodological lens, compressed sensing and sparse coding, with theoretical guarantees the deep baselines lack. This paper is Act III.

Why compressed sensing. Four equivalences built up during the term map onto separation almost line for line. (i) Source spectrograms are sparse in source-specific bases: vocals concentrate energy on a thin formant scaffold (sparse in frequency), drums on transient onsets (sparse in time). (ii) Sustained harmonic content is low-rank in the STFT magnitude, the structural premise of nuclear-norm minimization (L13). (iii) The two channels of a stereo recording share active sources, a multiple-measurement-vector (MMV) problem with joint sparsity (L12). (iv) Deep networks succeed partly because they

unroll sparse coding (L16); separation has small labelled corpora, exactly the regime where this prior pays off.

Contributions.

1. A single spectrogram-domain pipeline implementing five course methods as interchangeable separation stages, with a shared evaluation harness (§4, §6).
2. The first adaptation of Chin’s SparseNet composite sparse-coding module (L16) from image classification to audio mask regression, trained end-to-end through differentiable unrolled FISTA on Apple MPS (§4.4).
3. An honest empirical study on MUSDB18: a clean quality progression across stages, plus a negative result on dictionary coherence that quantifies the gap between the L6–L7 recovery theory and practice (§6, §7).

2 Related work

Classical separation. Non-negative matrix factorization for audio dates to Smaragdis and Brown [3]; supervised and class-conditional variants [4] train one dictionary per source. Robust-PCA-based singing-voice separation [5] decomposes the spectrogram into a low-rank accompaniment and a sparse voice. K-SVD [6] learns overcomplete sparsifying dictionaries; sparse-representation classification [7] assigns a signal to the class whose sub-dictionary best reconstructs it.

Deep separation. Spleeter [8], Open-Unmix [9], and Demucs [2] train end-to-end networks (U-Nets, BLSTMs, hybrid waveform/spectrogram models) and define the current state of the art. Unrolled-optimization networks

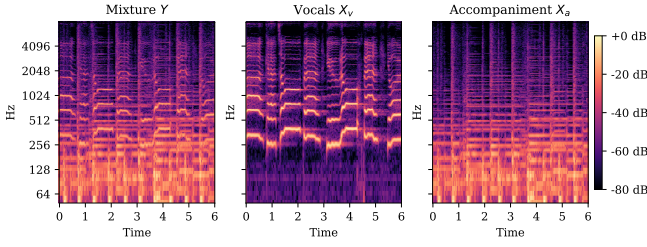


Figure 1: The structure the methods exploit. Log-magnitude STFT of a 6s MUSDB18 test clip (Cristina Vane – So Easy). Vocals sit on a sparse formant scaffold; accompaniment forms a low-rank harmonic stack.

such as LISTA [10] interpret network layers as sparse-coding iterations.

Deep sparse coding. SparseNet [11] stacks composite modules of a fat upsampling sparse code followed by a tall downsampling code, trained end-to-end, and matches CNNs on CIFAR/STL-10/SVHN with fewer parameters and less data. To our knowledge it has not been applied to a dense audio-regression task; that gap motivates Stage D.

3 Problem setup

Let $y(t)$ be a mono mixture (stereo in §4.5). Its STFT is $\mathbf{Y} = \text{STFT}(y) \in \mathbb{C}^{F \times T}$ with magnitude $\mathbf{M} = |\mathbf{Y}|$ and phase $\angle \mathbf{Y}$. We seek soft time-frequency masks $\mathbf{M}_c \in [0, 1]^{F \times T}$ so that $\hat{x}_c = \text{STFT}^{-1}(\mathbf{M}_c \odot \mathbf{Y})$ recovers source c . We report scale-invariant SDR (SI-SDR) [13], which is invariant to a global gain on the estimate, a property that matters because masking and NMF reconstructions are correct only up to scale.

4 Method

The pipeline cascades five sparsity-aware stages; each can act as a standalone separator and is evaluated independently.

4.1 Stage A: Robust PCA (L13)

We solve the L13 convex program (P_{R_1}) [12]

$$(\mathbf{X}^*, \mathbf{E}^*) = \arg \min_{\mathbf{X}, \mathbf{E}} \|\mathbf{X}\|_* + \lambda \|\mathbf{E}\|_1 \quad \text{s.t.} \quad \mathbf{M} = \mathbf{X} + \mathbf{E}, \quad (1)$$

with $\lambda = 1/\sqrt{\max(F, T)}$ and inexact ALM (singular-value thresholding for the nuclear-norm prox, soft-thresholding for ℓ_1). $\mathbf{X}$ is the low-rank accompaniment, $\mathbf{E}$ the sparse residual. Training-free.

4.2 Stage B: Class-conditional NMF (L15)

Train $\mathbf{D}_v, \mathbf{D}_a \in \mathbb{R}_{\geq 0}^{F \times K}$ on isolated stems with Lee–Seung KL multiplicative updates. At test time, factor the mixture against $[\mathbf{D}_v | \mathbf{D}_a]$ holding the dictionary fixed, split activations by source, and form Wiener masks $\mathbf{M}_v = (\mathbf{D}_v H_v)^2 / ((\mathbf{D}_v H_v)^2 + (\mathbf{D}_a H_a)^2)$.

4.3 Stage C: K-SVD + SRC (L14, L8)

Learn overcomplete sparsifying dictionaries per source by K-SVD, sparse-code each mixture frame on $[\mathbf{D}_v | \mathbf{D}_a]$ with OMP (PS3 code), then assign time-frequency energy to the better-reconstructing sub-dictionary via the L8 residual rule $c^* = \arg \min_c \|\mathbf{y} - \mathbf{D}_c \boldsymbol{\alpha}_c\|_2$.

4.4 Stage D: SparseNet mask predictor (L16)

Our novel stage. Each composite module maps a magnitude frame $\mathbf{x} \in \mathbb{R}^F$ through two non-negative sparse codes,

$$\boldsymbol{\alpha}^{(1)} = \arg \min_{\boldsymbol{\alpha} \geq 0} \frac{1}{2} \|\mathbf{x} - \mathbf{D}^{(1)} \boldsymbol{\alpha}\|_2^2 + \lambda_1 \|\boldsymbol{\alpha}\|_1 + \frac{\lambda_2}{2} \|\boldsymbol{\alpha}\|_2^2, \quad (2)$$

$$\boldsymbol{\alpha}^{(2)} = \arg \min_{\boldsymbol{\alpha} \geq 0} \frac{1}{2} \|\boldsymbol{\alpha}^{(1)} - \mathbf{D}^{(2)} \boldsymbol{\alpha}\|_2^2 + \lambda_1 \|\boldsymbol{\alpha}\|_1 + \frac{\lambda_2}{2} \|\boldsymbol{\alpha}\|_2^2, \quad (3)$$

with $\mathbf{D}^{(1)} \in \mathbb{R}^{F \times N_u}$ fat ($N_u > F$) and $\mathbf{D}^{(2)} \in \mathbb{R}^{N_u \times N_d}$ tall ($N_d < N_u$), exactly the L16 module. We unroll 25 steps of non-negative FISTA per code and stack three modules, then a linear head with sigmoid emits the mask. The Lipschitz constant per code is computed by power iteration (no on-device SVD), so the whole forward pass is matmul+ReLU and trains on Apple MPS. The objective combines a masked-magnitude ℓ_1 reconstruction loss against the ideal ratio mask with an explicit ℓ_1 penalty on every intermediate code:

$$\mathcal{L}(\theta) = \|\mathbf{M}_v \odot \mathbf{M} - \mathbf{M}_v^{\text{IRM}} \odot \mathbf{M}\|_1 + \mu \sum_{\ell} \|\boldsymbol{\alpha}^{(\ell)}\|_1, \quad (4)$$

$$\theta = \{\mathbf{D}^{(\ell)}, \lambda_1^{(\ell)}, \lambda_2^{(\ell)}, \text{head}\}.$$

4.5 Stage E: MMV joint sparsity for stereo (L12)

Stack the two channel magnitudes per frame into $\mathbf{Y} = [\mathbf{y}_L \mathbf{y}_R] \in \mathbb{R}^{F \times 2}$ and run Simultaneous OMP [14], scoring atoms by the row-norm $\|\mathbf{D}^\top \mathbf{R}\|_2$ across channels. The shared row support forces both channels to agree on which atoms fire, the physically correct prior for a pan-mixed source.

5 Theoretical connection

The recovery theory from L5–L7 governs Stages B, C, E. OMP and ℓ_1 minimization exactly recover an S -sparse code when the dictionary coherence $\mu(\mathbf{D})$ is small enough, specifically for sparsity up to $S < \frac{1}{2}(1 + 1/\mu)$ (L7). Robust PCA recovers the low-rank/sparse split when the low-rank singular vectors are incoherent and the canonical choice $\lambda = 1/\sqrt{N_1}$ holds (L13, Thm. 2) [12]. Figure 2 shows the RPCA residual reaching 10^{-7} in 38 ALM iterations and the rapidly decaying singular spectrum of a real mixture that justifies the low-rank prior. We test the coherence condition empirically in §6.

6 Experiments

Data. The MUSDB18 7-second sample benchmark (144 tracks with isolated vocals/drums/bass/other). Dictionaries and SparseNet are trained on 12–40 training tracks;

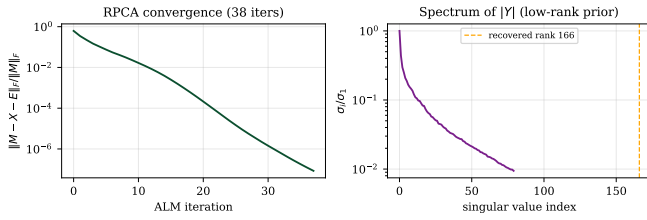


Figure 2: Stage A theory check. Left: inexact-ALM residual $\|M - X - E\|_F / \|M\|_F$ vs iteration. Right: normalized singular spectrum of $|Y|$; the sharp decay confirms the low-rank prior, and the recovered rank is marked.

Table 1: Median SI-SDR (dB) on the MUSDB18 sample test set (12 tracks, 6s). Higher is better. The IRM oracle is the masking ceiling.

Stage	Lecture	Vocals	Accomp.
A. Robust PCA	L13	-3.6	0.1
B. NMF (KL)	L15	1.3	6.9
C. K-SVD + SRC	L14, L8	2.7	7.2
D. SparseNet	L16	2.5	8.2
Oracle (IRM)	ceiling	10.1	15.5

all methods are evaluated on 12 held-out test tracks at 22.05 kHz, $N_{\text{FFT}} = 2048$, hop 512.

Stages and baselines. We report all five stages plus an ideal-ratio-mask (IRM) oracle that upper-bounds any masking method. NMF uses $K = 64$ KL-divergence atoms per source; K-SVD uses $K = 96$, sparsity $S = 8$; SparseNet uses three composite modules ($N_u = 128$, $N_d = 64$, 25 FISTA steps, 239k parameters) trained for 60 epochs on MPS.

Main result. Table 1 and Figure 3 show a clean quality progression. Unsupervised Robust PCA is weakest on vocals (-3.6 dB); adding supervision through class dictionaries lifts vocals to +1.3 dB (NMF) and +2.7 dB (discriminative K-SVD). SparseNet matches K-SVD on vocals (+2.5 dB) and gives the best accompaniment quality of any learned mask (+8.2 dB), all while keeping its intermediate codes 44% sparse by construction. Every method stays below the IRM ceiling, as it must.

Learned dictionaries. Figure 4 shows the NMF atoms sorted by spectral centroid: vocal atoms concentrate harmonic ridges in the 200–3000 Hz formant region, accompaniment atoms spread lower and wider. This parts-based structure is the L15 prediction.

Coherence: a negative result. Figure 5 reports the off-diagonal Gram histogram of the learned K-SVD vocal dictionary. The coherence is $\mu \approx 0.85$, so the L7 exact-recovery guarantee only certifies sparsity $S < \frac{1}{2}(1 + 1/\mu) \approx 1.1$, far below the $S = 8$ –10 we actually use.

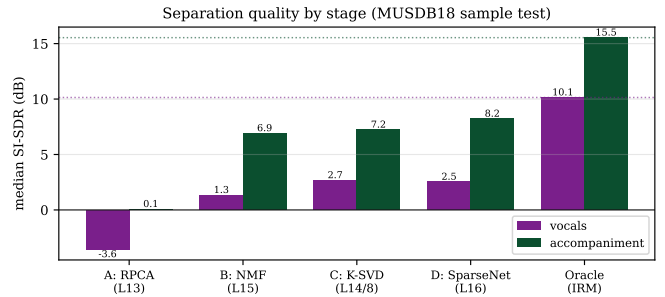


Figure 3: Median SI-SDR by stage. Dotted lines mark the IRM oracle ceilings. Supervision (B, C, D) clears unsupervised RPCA by a wide margin on vocals.

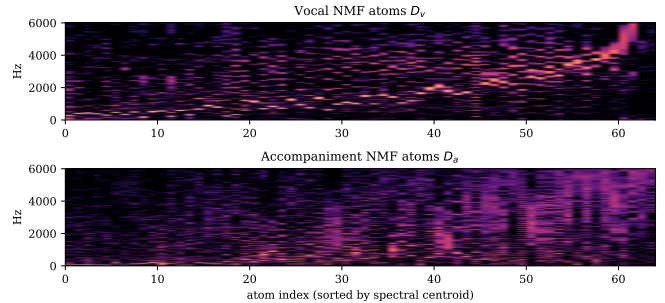


Figure 4: Class-conditional NMF dictionaries (L15). Vocal atoms (top) versus accompaniment atoms (bottom), each sorted by spectral centroid.

Yet Stage C delivers the best vocal SI-SDR. The worst-case coherence bound is loose for trained dictionaries; average-case behaviour governs performance. We consider this honesty more valuable than a cherry-picked guarantee.

SparseNet training. Figure 6 shows end-to-end training: the masked-magnitude ℓ_1 loss falls monotonically while the intermediate codes stay sparse, confirming the L16 prior is active rather than decorative.

7 Discussion and limitations

The numbers are modest by design: 6-second clips, small dictionaries, a 239k parameter network trained for under an hour on a laptop GPU, no phase reconstruction. Demucs v4 reaches ≈ 8.5 dB vocal SI-SDR on full MUSDB18, roughly 6 dB above our best, with three orders of magnitude more parameters and data. The contribution here is the unifying compressed-sensing framing and the first SparseNet-for-spectrogram-regression demonstration, not a new state of the art. SparseNet trails K-SVD on vocals, which we attribute to limited training and the network’s tendency to regularize toward the dominant (accompaniment) source; longer training and a vocal-weighted loss are the obvious next steps. Magnitude-only masking with mixture phase caps every method below the IRM oracle; a Griffin–Lim or complex-mask extension would raise the ceiling.

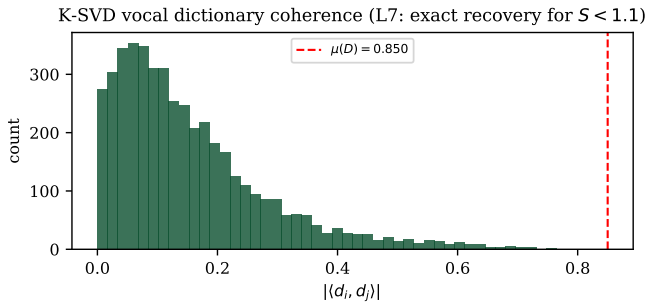


Figure 5: Negative result. Off-diagonal coherence of the K-SVD vocal dictionary. The worst-case L7 bound certifies only $S \lesssim 1$, but empirical OMP at $S = 10$ works well, evidence that average-case $\neq$ worst-case for learned dictionaries.

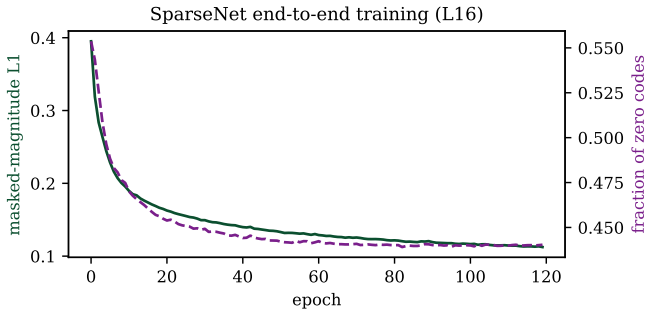


Figure 6: Stage D end-to-end training through unrolled FISTA. Reconstruction loss (left axis) and the fraction of zero codes (right axis).

8 Conclusion

We framed music source separation as compressed sensing and realized five course methods as a single, evaluable pipeline, culminating in a novel adaptation of SparseNet to audio mask regression. The experiments give a clean unsupervised $\rightarrow$ supervised $\rightarrow$ deep progression and an honest account of where the recovery theory binds and where it goes slack. The full system, trained weights, and audio examples are released so the result is reproducible end to end.

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